M. Ikram Ullah

AI/ML Engineer · LLM Agents, RAG & AI Products

Islamabad, Pakistan | (+92) 311-7862537 | ikramullah4748@gmail.com | linkedin.com/in/ikram-ullah11 | github.com/ikram4748 | ikram4748.github.io

SUMMARY

AI/ML engineer with 3+ years of building LLM agents, RAG systems and AI products that real people use, including VSARA, four connected healthcare apps on Google Play with LLM chat, voice and agent workflows. 30+ projects for clients and my own portfolio, and I measure what I build: a self-correcting text-to-SQL agent (+6.0 points on held-out BIRD questions), a support agent with guardrails in code (0 policy violations vs 3 for a prompt-only agent), and a RAG benchmark where HyDE raised answer accuracy from 47% to 61%. I work across Python/FastAPI and Django backends, React Native/TypeScript frontends and LLM APIs; pursuing an MS in Data Science.

TECHNICAL SKILLS

LLM agents & RAG: LangGraph (state machines, interrupts, checkpointers), LangChain, CrewAI, function calling, multi-agent pipelines, hybrid retrieval (BM25 + dense, RRF), cross-encoder reranking, HyDE, Qdrant, FAISS, ChromaDB
Evaluation & safety: execution accuracy, RAGAS-style faithfulness, LLM-as-judge with validation, simulated-user (tau-bench style) evaluation, dev/test splits, paired bootstrap CIs, prompt-injection defences, guardrails
Models & APIs: Gemini, OpenAI and Anthropic APIs, Hugging Face Transformers, sentence-transformers, BGE-M3, ONNX Runtime / fastembed, Whisper ASR, Azure AI Speech, open-weight LLMs (Qwen, Gemma, Llama)
ML & deep learning: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, LightGBM, CNNs, Vision Transformers, transfer learning, YOLOv8, Grad-CAM, SHAP
Engineering: Python, TypeScript, SQL, FastAPI, Django REST, React, React Native (Expo), Supabase, PostgreSQL, Docker, Git, pytest, Streamlit, n8n

EXPERIENCE

AI Engineer & Mobile App Developer, Syslab Technologies *(Islamabad, Pakistan)* Oct 2024 – Nov 2025

- Built and shipped VSARA, a healthcare platform of four connected React Native (Expo) apps on Google Play: SARA Life (patients), SARA Care (doctors), SARA Pharmacy and Flexa, with LLM-powered conversational interfaces and Azure AI Speech voice interaction.
- Architected the platform's agentic voice and chat assistant, combining a RAG pipeline (FAISS + Hugging Face sentence-transformers) with Whisper speech recognition and neural text-to-speech.
- Built n8n automation workflows orchestrating multi-step agent actions (appointment scheduling, symptom-triage routing, EHR lookup), improving response throughput by 31%.
- Improved diagnostic response accuracy by 23% through systematic A/B testing of model configurations, prompt strategies and RAG retrieval settings (chunk size, retrieval depth).
- Built an automated model-monitoring pipeline (accuracy, precision, recall, F1, latency) with real-time Power BI dashboards for early detection of model drift.
- Investigated model failures through error logs, edge cases and interaction data; targeted retraining and data-quality fixes reduced the error rate by 18%.

Freelance AI Engineer & Mobile App Developer, Self-employed (direct clients) *(Remote)* Feb 2023 – Present

- Delivered 15+ end-to-end AI/ML projects for healthcare, e-commerce and fintech clients, from scoping and data collection to deployment and handoff.
- Built and published JSuiteCloud, an ERP app on the Apple App Store (React Native, Expo, TypeScript), working inside an existing production codebase; now building an AI-powered children's learning platform with LLM conversational feedback and audio flows (React Native, TypeScript, Django REST).
- Built RAG chatbots and document Q&A systems (LangChain, FAISS, Hugging Face embeddings) that answer from clients' internal knowledge bases with cited, grounded responses.

PROJECTS

Customer-Ops Agent with Guardrails *(github.com/ikram4748/customer-ops-agent)*
LangGraph · Gemini function calling · SQLite · Streamlit

- Built a support agent that executes refunds, cancellations and address changes behind a gateway the model cannot bypass: identity and argument-provenance checks, policy enforced in code, customer confirmation codes, supervisor approval through LangGraph interrupts, and idempotency keys that stop double refunds when a payment times out.

- Designed a tau-bench-style evaluation of 45 simulated-customer scenarios (prompt injection, social engineering, injected tool failures) scored on the final database state: on 30 held-out scenarios the guarded agent reached 100% task success with 0 policy violations, vs 90% and 3 violations (including a double refund after a payment timeout) for a prompt-only agent on the same model.

Data Analyst Agent (text-to-SQL) *(github.com/ikram4748/data-analyst-agent)*
LangGraph · Gemini · sqlglot · DuckDB · fastembed · Streamlit

- Built a self-correcting text-to-SQL agent: local-embedding schema linking, a two-layer read-only SQL guard, a literal-value check that catches silent zero-result errors, and a bounded fix loop driven by database feedback.
- Raised execution accuracy on 150 held-out BIRD questions from 60.7% (one-shot baseline, same model) to 66.7% (+6.0 points, 95% CI +1.3 to +11.3) at 1.09 LLM calls per question; tuned on a separate dev split after error analysis of 48 failures.
- Found and fixed a measurement bug in which rate-limit fallback silently mixed three models across benchmark variants.

Literature Review Agents *(github.com/ikram4748/research-assistant-agent)*
LangGraph · Gemini · arXiv & OpenAlex APIs · fastembed

- Built a multi-agent reviewer (planner, searcher, reader, writer, critic, reviser) in which every finding must quote its source abstract word for word, checked in code, and a critic sends unsupported sentences back for revision.
- On 20 held-out questions both systems reached 97% sentence support; multi-agent cited more precisely (87% vs 80%), but the paired 95% CI included zero at 6× the LLM calls. Fixed a judge that passed every sentence: its catch rate for deliberately wrong citations rose from 48% to 90%.

Financial RAG Lab *(github.com/ikram4748/financial-rag-lab)*
PyMuPDF · BM25 · bge-small · cross-encoders · Gemini

- Built a RAG benchmark on FinanceBench (150 analyst questions, 84 filings, 12,013 pages) comparing 3 chunking strategies, 3 metadata scopes and 5 retrievers (BM25, dense, hybrid RRF, two cross-encoder rerankers) plus HyDE, scored on evidence-page recall with a dev/test split.
- On 100 held-out questions, dense retrieval with company/year filtering beat BM25, hybrid and cross-encoder reranking; HyDE raised evidence recall@5 from 44% to 60% and answer accuracy from 47% to 61% (+14 points, 95% CI +5 to +23), vs 24% for naive RAG and 84% with the evidence pages given, which showed retrieval was the bottleneck. Answers graded by a separate judge model validated on gold and corrupted answers.

Qanoon Assist: legal-information RAG for Pakistani law (private) *(code available on request)*
FastAPI · React · PostgreSQL · Qdrant · BGE-M3 · Gemini

- Scraped and parsed 6,664 federal and provincial laws (117,365 sections, 147,701 chunks) from five government portals; answers in English, Urdu and Roman Urdu with section-level citations.
- Hybrid BM25 + BGE-M3 dense retrieval with bge-reranker-v2-m3 and citation verification; hit@5 1.00, faithfulness 0.944 and 4.8 s average latency on a 36-question evaluation; 124 automated API and UI tests.

CLIENT PROJECTS (CODE PRIVATE)

- **AI Email Template Generator (mobile app)** *(React Native · OpenAI GPT-4o-mini)*: Mobile app that drafts editable e-mails in 8+ categories and four tones.
- **Multi-Agent Content Team** *(CrewAI · web search)*: A researcher agent with live web search hands its findings to a writer agent, so briefs come from real research.
- **Autonomous Data Analyst** *(Gemini · pandas · sandboxed exec)*: Answers plain-English questions about any spreadsheet by having the LLM write pandas code that runs in a restricted sandbox; shows the computed result and the code.
- **MedVision (chest X-ray classification)** *(TensorFlow/Keras · ResNet50 · EfficientNetB0 · Grad-CAM)*: Pneumonia detection on 5,856 clinical X-rays comparing a CNN from scratch, frozen ResNet50 and fine-tuned EfficientNetB0; Grad-CAM heatmaps show which lung regions drive each prediction.

RESEARCH

UrduComicLab: complete-story Urdu story-to-comic generation (MS thesis, in progress)

- Open-weight pipeline turning complete Urdu stories into right-to-left comics; vs baseline: spoken lines printed 0.49 → 1.00, dialogue F1 0.63 → 1.00, face consistency 0.54 → 0.67 (paired bootstrap).

EDUCATION

MS Data Science, Quaid-e-Azam University, Islamabad — Feb 2025 – Present
Coursework: Machine Learning, Deep Learning, NLP, Big Data Analytics, Statistical Modeling, Data Visualization. Thesis: UrduComicLab.